

SATNET EDU / Lecture 05

Designing and Evaluating Satellite Networks

How to turn a network idea into a controlled, reproducible experiment.

LECTURE 05

From a design question to an explained result

State the question. Control the comparison. Inspect the outcome.

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Start with a Question That the Model Can Answer

Connect a design choice to a measurable outcome under stated conditions.

► **Turn a goal into a comparison.**

- Ask how a specific topology change affects connectivity between two chosen ground sites.

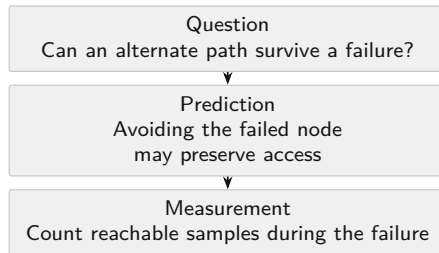
► **Predict a mechanism first.**

- Allowing more candidate ISLs may create alternate paths. Range and Earth blockage can still remove those links.

► **Match the claim to the model.**

- SatNet Edu can compare geometric reachability and propagation. It cannot predict throughput without a traffic model.

Make the question testable



Fix endpoints, time window, geometry, and routing metric

The teaching model omits antenna scheduling, capacity, queues, packet loss, and routing convergence.

Change One Factor and Keep the Rest Comparable

A controlled comparison helps identify why a measured result changed.

► **Use a shared baseline.**

- Use the same epoch, constellation, sites, elevation mask, duration, and sample times in both runs.

► **Name the factor you change.**

- Here the failure schedule is the treatment. Add one satellite outage while leaving geometry and routing settings fixed.

► **Separate follow-up questions.**

- A later experiment can change topology or altitude. Changing both at once makes a single-cause explanation harder.

The paired failure experiment

Setting	Baseline	Failure run
Satellites	6×12	6×12
Altitude / tilt	1000 km / 53°	1000 km / 53°
Access mask	10°	10°
ISL limit	4000 km	4000 km
Window / step	20 min / 10 s	20 min / 10 s
Outage	None	600–900 s

Vancouver $\longleftrightarrow$ Tokyo Same epoch

Companion experiment uses orbital-neighbor links and epoch 2026-01-01 UTC. The node selection rule is documented.

Coverage, Reachability, and Delay Need Different Counts

Always state the population used in a metric, especially when some samples have no route.

► **Coverage is local to one site.**

- Count samples with at least one usable ground link. This does not require a route to the destination.

► **Reachability is a pairwise property.**

- Count samples with a complete source-to-destination path, using all scheduled observation times as the denominator.

► **Delay is conditional on a route.**

- Average valid delays only and report the reachability fraction alongside them. Missing paths are not zero-delay paths.

The denominator changes the meaning

Time	U access	U-V route	Delay
0 s	Yes	Yes	12 ms
10 s	Yes	No	—
20 s	Yes	Yes	18 ms
30 s	No	No	—

Coverage at U: $3/4 = 75\%$

Pair reachability: $2/4 = 50\%$

Conditional delay: $(12 + 18)/2 = 15$ ms

Constructed four-sample example. Delay here means one-way propagation. These are sample fractions, not exact time fractions.

Unequal Sample Spacing Requires Time Weights

Extra samples near an event can change a sample average without changing the time history.

► **Samples can represent unequal time.**

- A recorder can add samples at failure times. Counting every observation equally can overweight short periods.

► **Choose and label an estimator.**

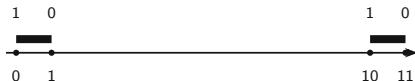
- A left-hold estimate assumes the observed state lasts until the next sample. Weight each state by that interval length.

► **Check sensitivity to finer sampling.**

- This assumption can miss an unobserved change. Report the interval, window, and approximation with the metric.

Two reachable samples, unequal durations

a : 1 for reachable, 0 otherwise



Sample fraction: $2/4 = 50\%$

Held-state time fraction: $(1 + 1)/11 = 18.2\%$

Time in seconds. Final sample has zero following duration.

Constructed states at 0, 1, 10, and 11 s. The time fraction assumes the pictured state is held between samples.

A Failure Test Should Target a Defined Component

A failure experiment is meaningful only when the disabled component and its timing are specified.

► **Choose a repeatable selection rule.**

- Select an interior satellite on the baseline route at 600 seconds. Save its ID so the stress case can be reproduced.

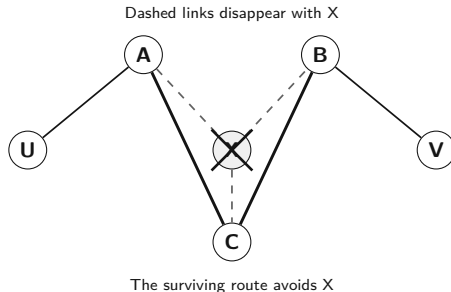
► **Remove every incident link.**

- A disabled satellite cannot forward traffic. Removing just one displayed route edge would test a different fault.

► **Look for a surviving alternate path.**

- The replacement can be longer, or no path may remain. Neither outcome should be assumed before querying the graph.

One failed node affects multiple links



Constructed graph. A targeted route-node failure tests a stress case, not average fleet reliability.

Compare the Failed Run with Its Matched Baseline

Orbital motion continues during a failure, so before-and-after values alone can mix two effects.

► Compare the same timestamps.

- At each timestamp, compare both runs under identical motion. Their difference isolates the scheduled failure.

► Separate outage from extra delay.

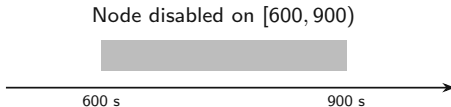
- Calculate delay differences only where both runs have a route. Report additional unreachable samples separately.

► Inspect recovery independently.

- Restoring a node allows links to return. The model recomputes routes immediately, without protocol recovery delay.

A matched comparison during one event

Time	Baseline	Failure	Difference
590 s	12 ms	12 ms	0 ms
600 s	13 ms	18 ms	+5 ms
700 s	14 ms	No path	—
900 s	15 ms	15 ms	0 ms



Constructed comparison values. The companion script supplies actual traces and metrics for the chosen scenario.

Keep Designs That Offer a Useful Tradeoff

No single metric captures satellite count, connectivity, and path quality at the same time.

► **Choose a small set of objectives.**

- Here we prefer fewer satellites, a higher reachable-sample fraction, and lower conditional propagation delay.

► **Reject a dominated design.**

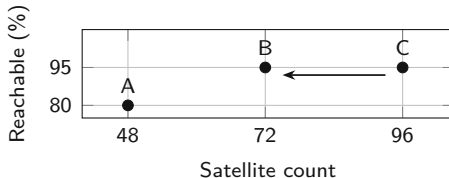
- B beats C on satellite count and delay while matching reachability. C offers no benefit under these chosen metrics.

► **Keep useful alternatives.**

- A is cheaper but less connected than B. Choosing between them requires an explicit service goal or resource budget.

An illustrative three-design comparison

Design	Satellites	Reachable	Delay
A	48	80%	24 ms
B	72	95%	20 ms
C	96	95%	25 ms



Constructed results, not simulator output. Satellite count is a resource proxy, not a complete financial cost.

Save the Scenario and the Recorded Outcomes

A reproducible result needs more than a screenshot of a favorable instant.

► **Preserve inputs and selection rules.**

- Record the epoch, orbit parameters, topology, endpoints, failure ID, and sampling schedule with the script.

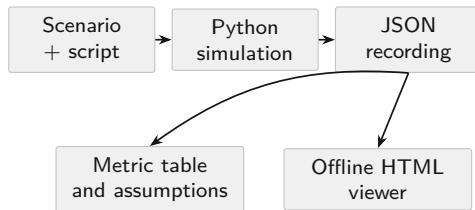
► **Keep the complete recording.**

- Save states and routes, including no-path samples. A viewer can replay the recording without running Python again.

► **State the model and time window.**

- Include dependency versions, metrics, and limitations. A short synthetic run does not establish worldwide operational service.

A result another student can inspect



The viewer displays records. It does not recompute routes.

The exported trace includes scenario data and dependency versions. The basic model is deterministic.

Build a Defensible Satellite-Network Experiment

Combine architecture, coverage, routing, and time-varying behavior in a small final investigation.

► **Begin with the paired failure script.**

- Predict the outcome, run both cases, and locate any extra disconnections or detours in the viewer.

► **Make one justified design revision.**

- Change the topology policy or constellation layout. Repeat the same failure rule and compare the same metrics and sites.

► **Explain a result and a limitation.**

- Use a route diagram and a time plot to explain the mechanism. State what the experiment cannot conclude.

Use the same failure ID to isolate a design change. Retargeting a new route node asks a different stress-test question.

A compact final submission

Include	Answer
Question + prediction	What should change, and why?
Baseline + revision	Which design factor changed?
Saved JSON + HTML	Can another student inspect it?
Metrics + time plot	How often, how long, how far?
Conclusion + limits	What does this run support?

```
python teaching/lecture-05/experiment.py
```

Outputs: paired traces, viewers, and CSV metrics